

When Similarity Isn't Enough: Zero-Shot Classifiers & Reasoning

Rachit Agarwal (230820)
rachitag23@iitk.ac.in
Indian Institute of Technology Kanpur

Abstract—Growing demand for efficient alternatives to ever-larger LLMs has drawn attention to lightweight models such as Jev and Laya, which make a bounded decision over a fixed set of options in a single pass instead of generating text. We study bi-encoders and cross-encoders, which score a fixed label set at lower cost than an LLM but whose captured information is poorly understood. We examine where semantic matching stops being sufficient for zero-shot classification. Both families perform well when labels are recoverable from semantic evidence but are substantially weaker on CSQA, where plausible candidates are poorly separated. Adding LLM-generated rationales raises accuracy and improves the standardized margin for all ten scorers. These results suggest that part of the gap may arise from constructing useful intermediate information rather than only recognizing its consequence. All code for this work is available at <https://github.com/rachit6105/jev-lite>.

Index Terms—zero-shot classification, bi-encoder, cross-encoder, natural language inference, commonsense reasoning, large language models.

I. INTRODUCTION

Many practical prediction problems do not require free-form generation. The output may already be known to lie in a fixed set of classes, intents, actions, or answer choices. In such settings, generating a textual response can be unnecessarily expensive, and recent work such as Jev [1] and Laya [2] has renewed interest in models that make bounded decisions directly. This motivates a broader question for efficient inference: how far can fixed-compute decision models go before the task requires information that is not already recoverable from the input representation?

Zero-shot classification provides a useful testbed because it reduces prediction to choosing from a fixed candidate set without task-specific training. We study bi-encoders and cross-encoders as accessible fixed-compute scorers that are conceptually close to the bounded-decision setting motivated by Laya and Jev, without attempting to reproduce either system.

The key limitation appears when a task requires implicit or intermediate information rather than a direct decision from available semantic evidence. Cheap bounded scorers may therefore be effective for direct-decision problems without being substitutes for generative models on tasks that require constructing such information.

We evaluate this boundary on AG News, Banking77, Emotion, and CommonsenseQA, spanning topic recognition, fine-grained intent discrimination, contextual affect

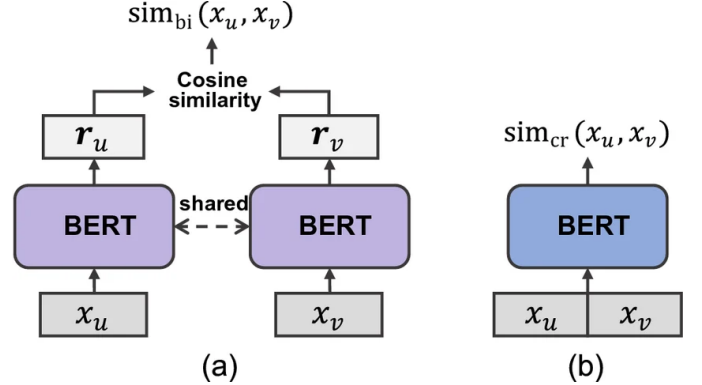


Fig. 1: Abstraction model for (a) bi-encoder and (b) cross-encoder architectures.

recognition, and commonsense inference. We also test class descriptions, task-specific fine-tuning, and rationale-assisted CSQA to distinguish failures of candidate scoring from failures to construct the information needed for that scoring.

II. BACKGROUND AND RELATED WORK

BERT-style transformers are the standard baseline for text classification: an encoder processes the input and a task-specific linear head maps it to logits over a fixed class set. This is accurate but needs supervised fine-tuning for every new task.

A. Bi-Encoders

To enable flexible, open-vocabulary classification, bi-encoders recast classification as representation matching, as in Sentence-BERT [3]. Recent methods improve zero-shot accuracy by turning labels into prompts and retrieving external text to enrich them [4]. Because label embeddings can be pre-computed, bi-encoders scale highly efficiently.

B. Cross-Encoders

Cross-encoders are the interaction-based alternative. Yin et al. and Laurer et al. cast zero-shot classification as NLI [5], [6]. Class descriptions ground the label further [7], but performance varies widely across datasets and often relies on lexical overlap and template artifacts [8]. Rerankers such as monoBERT [9] are the relevance-scoring variant.

C. Middle-Ground Architectures and Distillation

Intermediate architectures such as poly-encoders, ColBERT, and cross-encoder-to-bi-encoder distillation trade interaction capacity for cheaper retrieval-style inference [10], [11], [12].

D. Decision Models, LLMs, and the Reasoning Gap

Reasoning benchmarks resist surface matching and need world knowledge and relational deduction. Decision models such as Jev [1] and Laya [2] package this fixed-set setting directly: instead of generating text, they return a bounded answer (a choice, score, or probability) over options defined in advance, with fixed compute per query. Generative Large Language Models (LLMs) instead spend intermediate computation on each reasoning step, through explicit reasoning chains [13], latent reasoning [14], or an encode-think-decode process [15]. Fixed-set scorers collapse this into a bounded pass over each candidate. Comparing the two shows whether reasoning deficits come from collapsed computation, model scale, or the scoring objective itself. We do not evaluate Jev or Laya; our bi-encoders and cross-encoders are open, fixed-compute stand-ins for this model class.

III. THEORY AND MATHEMATICAL FORMULATION

This section defines the zero-shot scorers evaluated here: bi-encoders, NLI cross-encoders, and relevance rerankers (LLMs serve only as a conceptual reference). All are BERT-style transformers. They differ in size, in how text and label are combined, and in how the output becomes a score.

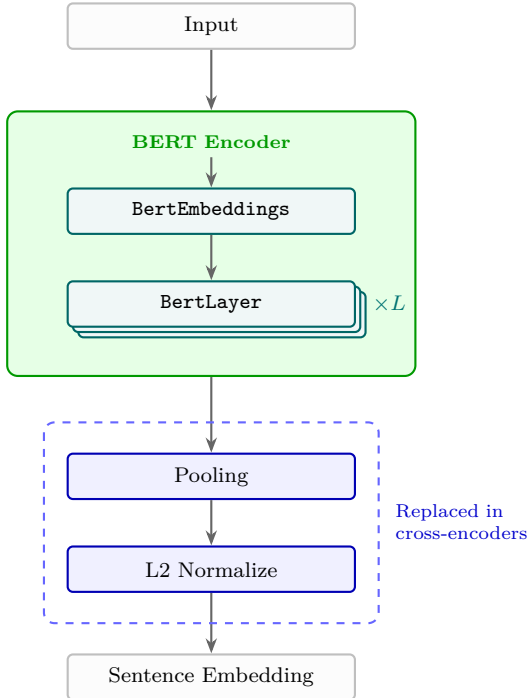


Fig. 2: BERT-style skeleton as used by the bi-encoders. Cross-encoders replace the output network with a classification head.

A. General BERT Encoders

Given a text x and a fixed label set $\mathcal{Y} = \{y_1, \dots, y_K\}$, a zero-shot scorer assigns each label a score $s(x, y_k)$ and predicts

$$\hat{y} = \arg \max_{y_k \in \mathcal{Y}} s(x, y_k),$$

A candidate label y_k can be replaced by a natural-language description $d_k = D(y_k)$, where D maps each class name to a textual definition. The input is either y_k or d_k , never both. The scores become

$$s_{\text{bi}}^{\text{desc}}(x, y_k) = \cos(f_\theta(x), f_\theta(d_k))$$

$$s_{\text{NLI}}^{\text{desc}}(x, y_k) = l^{(\text{ent})}(g_\theta(x, T(d_k)))$$

where $s_{\text{bi}}^{\text{desc}}(x, y_k)$ and $s_{\text{NLI}}^{\text{desc}}(x, y_k)$ are scores for bi-encoder and NLI models using description of labels.

Every model follows the same skeleton (Fig. 2):

- input network: token embedding, position, Layer-Norm;
- central block: self-attention and feed-forward layers, repeated L times;
- output network: pooling or a classification head.

The models differ in central block design, dimensions, and training method.

To turn scores into probabilities we use the softmax function, which maps a vector $\mathbf{z} \in \mathbb{R}^n$ to a distribution,

$$\text{softmax}(\mathbf{z}) = \frac{\exp(z_i)}{\sum_{j=1}^n \exp(z_j)},$$

and we write $\text{softmax}(\mathbf{z}/\tau)$ when a temperature $\tau > 0$ is applied.

B. Bi-Encoders

A bi-encoder independently maps the input text x and each candidate label y_k into a shared embedding space using the same encoder f_θ :

$$\mathbf{q} = f_\theta(x), \quad \mathbf{c}_k = f_\theta(y_k).$$

Text and label are each compressed into one vector before comparison, so there is no token-level interaction between them (the interaction bottleneck). The similarity is measured by cosine similarity and a temperature scale τ :

$$s_{\text{bi}}(x, y_k) = \frac{1}{\tau} \frac{\mathbf{q}^\top \mathbf{c}_k}{\|\mathbf{q}\|_2 \|\mathbf{c}_k\|_2}.$$

For L2-normalized embeddings, $\|\mathbf{q}\|_2 = \|\mathbf{c}_k\|_2 = 1$, this simplifies to

$$s_{\text{bi}}(x, y_k) = \frac{\mathbf{q}^\top \mathbf{c}_k}{\tau}.$$

The cosine term measures angular similarity and τ controls the sharpness of the distribution. We use $\tau = 0.01$ for zero-shot scoring and $\tau = 0.05$ for fine-tuning. With $\mathbf{s}_{\text{bi}}(x) = [s_{\text{bi}}(x, y_1), \dots, s_{\text{bi}}(x, y_K)]$, the probability over the candidate set is

$$p(y_k | x) = \text{softmax}(\mathbf{s}_{\text{bi}}(x))_k.$$

Since softmax and division by a positive τ are both monotonic,

$$\arg \max_k \text{softmax}(\mathbf{z}/\tau)_k = \arg \max_k z_k,$$

where $z_k = \mathbf{q}^\top \mathbf{c}_k / (\|\mathbf{q}\|_2 \|\mathbf{c}_k\|_2)$ is the cosine similarity. Temperature changes confidence but not candidate ranking, and therefore does not affect accuracy.

We evaluate four bi-encoders: all-MiniLM-L6-v2 [3], [16], e5-small-v2 [17], bge-small-en-v1.5 [18], and bge-m3 [19]. They follow the skeleton in Fig. 2 and differ in depth, width, pooling, and training procedure (Table I). Our training and scoring procedure is in Section IV.

TABLE I: Bi-encoders evaluated.

Model	L	d	Params	Pool	Training
MiniLM [20]	6	384	22M	mean	Contrastive (1B pairs)
e5-small [21]	12	384	33M	mean	Weakly-supervised contrastive + supervised fine-tuning
bge-small [22]	12	384	33M	CLS	RetroMAE, contrastive, instruction fine-tuning
bge-m3 [23]	24	1024	568M	CLS	Multilingual contrastive + distillation

C. Pooling

The encoder returns one vector per token, $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_n] \in \mathbb{R}^{n \times d}$, where $\mathbf{h}_1$ is the [CLS] state, and a mask $m_i \in \{0, 1\}$ marks real tokens ($m_i = 0$ for padding). Pooling reduces $\mathbf{H}$ to a single vector $\mathbf{q} \in \mathbb{R}^d$. Both variants used here are weighted sums of the token vectors, $\mathbf{q} = \mathbf{H}^\top \mathbf{w}$, differing only in the weights $\mathbf{w} \in \mathbb{R}^n$:

$$\begin{aligned} \mathbf{q}_{\text{CLS}} &= \mathbf{h}_1, & w_i &= \mathbb{I}[i = 1], \\ \mathbf{q}_{\text{mean}} &= \frac{\sum_{i=1}^n m_i \mathbf{h}_i}{\sum_{i=1}^n m_i}, & w_i &= \frac{m_i}{\sum_j m_j}. \end{aligned}$$

CLS pooling puts all weight on one position that self-attention has trained to summarize the sequence. Mean pooling weights every real token equally, and the mask keeps padding from diluting the result. The pooled vector is then ℓ_2 -normalized, $\mathbf{q} \leftarrow \mathbf{q} / \|\mathbf{q}\|_2$, and the same pooling is applied to texts and candidates. Pooling is part of the model definition: a different pooling at inference gives vectors outside the geometry the model was trained to produce (Table I).

D. NLI Cross-Encoders

An NLI cross-encoder treats the text x as a premise and maps each candidate label to a hypothesis $h_k = T(y_k)$, e.g. $T(y_k) = \text{“This text is about } y_k\text{.”}$. The pair is concatenated into one sequence and encoded jointly as $\mathbf{z}_k = g_\theta(x, h_k)$, so attention can flow between text and label tokens. The classifier head produces logits $\mathbf{l}_k = [l_k^{(1)}, \dots, l_k^{(C)}]$, where C is the number of NLI classes: three

for entailment, neutral, contradiction, or two for entailment versus non-entailment (binary zero-shot models). The score used for ranking is the entailment logit

$$s_{\text{NLI}}(x, y_k) = l_k^{(\text{ent})},$$

so that $\hat{y} = \arg \max_k s_{\text{NLI}}(x, y_k)$. This differs from the per-pair probability $P(\text{ent} \mid x, h_k) = \text{softmax}(\mathbf{l}_k)_{\text{ent}}$, which normalizes over the C classes of that pair, depends on the other class logits, and can therefore rank candidates differently.

We evaluate five NLI models (Table II). Four are encoder-only: DeBERTa-v3 [24] (disentangled attention over content and relative position) in base and large sizes, and the XLM-RoBERTa-based bge-m3 [19]. BART-large-MNLI [25] is encoder-decoder, and its classification head reads the final end-of-sequence decoder state (Fig. 3). The models also differ in output head (3-way or binary) and training data.

TABLE II: NLI models evaluated

Model	L	d	Params	Head	Training
BART-large-MNLI [26]	12+12	1024	407M	3-way	MNLI
DeBERTa-L-ZS [27]	24	1024	435M	binary	NLI + tasks
DeBERTa-B-ZS [28]	12	768	184M	binary	NLI + tasks
NLI-DeBERTa-B [29]	12	768	184M	3-way	SNLI, MNLI
bge-m3-ZS [30]	24	1024	568M	binary	NLI + tasks

* ZS indicates zeroshot-v2.0. Head denotes a 3-way (entailment/neutral/contradiction) or binary (entailment/not entailment) classification. “NLI + tasks” indicates NLI data plus classification data reformatted as NLI.

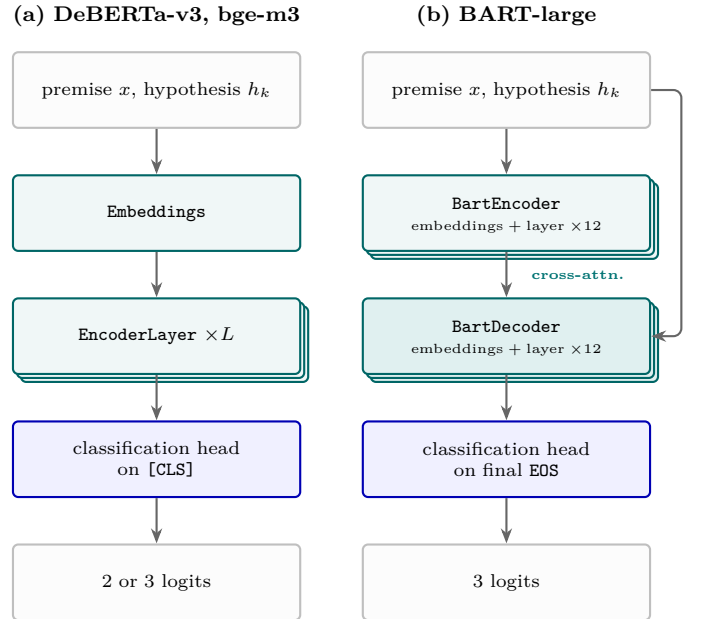


Fig. 3: NLI cross-encoder architectures: (a) encoder-only models, (b) BART, where the same sequence is fed to the encoder and the decoder.

E. Relevance Reranker

A reranker uses the same joint encoding as the NLI cross-encoders (Fig. 3a), $\mathbf{z}_k = g_\theta(x, y_k)$, but replaces the NLI head with a linear head that outputs one relevance logit,

$$s_{\text{rel}}(x, y_k) = \mathbf{w}^\top \mathbf{z}_k + b,$$

trained on query–passage relevance data [9]. Prediction remains $\hat{y} = \arg \max_k s_{\text{rel}}(x, y_k)$. It keeps full input–candidate interaction but measures topical relevance rather than directional entailment, and its scores are comparable across labels of the same text, not necessarily across texts. We evaluate bge-reranker-base [31], an XLM-RoBERTa-base model [32] with $L = 12$, $d = 768$, and about 278M parameters, trained with the C-Pack recipe [18].

F. Fine-Tuning Objective

For task-specific fine-tuning, let $\mathcal{D} = \{(x_i, y_i^*)\}_{i=1}^N$ and let $\mathbf{s}_i = [s(x_i, y_1), \dots, s(x_i, y_K)]$ be the unscaled scores (e.g., cosine similarities) for example i . The predicted distribution is

$$p_{ik} = \text{softmax}(\mathbf{s}_i / \tau)_k,$$

and the objective is the cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p_{i, y_i^*},$$

which raises the score of the correct candidate over its competitors. For bi-encoders this reshapes the shared embedding space so that $s_{\text{bi}}(x_i, y_i^*) > s_{\text{bi}}(x_i, y_j)$ for all $j \neq y_i^*$. Cross-encoders adjust the joint interaction function in the same way, with the softmax taken over the sampled candidate subset used in training (Section IV).

G. Computational Difference

Let E denote the cost of one transformer encoding and K the number of candidates. For a bi-encoder with cached label embeddings,

$$C_{\text{bi}} \approx E + O(K)$$

whereas for a cross-encoder

$$C_{\text{cross}} \approx O(KE)$$

because each pair (x, y_k) must be jointly encoded. Independent encoding is cheap; joint encoding gives more interaction at higher cost.

IV. METHODOLOGY

a) Input and Output: The input is a text x and a list of candidate strings (class names, class descriptions, or the five CommonsenseQA options). The output is the index of the highest-scoring candidate. Inputs are truncated to 512 tokens, and each model receives the formatting it was trained for: e5 gets **query:** on both text and candidates, whereas bge and MiniLM receive no prefix. The choice of

prefix is critical, as demonstrated by e5-small-v2 accuracy on AG News:

- **query:** / **query:** (default configuration): **68.5%**
- **query:** / **passage:** (mismatched candidates): **32.6%**
(a 35.9-point drop)

b) Scoring: Bi-encoders are scored as in Section III, with CLS pooling for bge and mean pooling for MiniLM and e5. Label embeddings are computed once per dataset, except for CSQA where the options change per question.

For NLI cross-encoders, the hypothesis is built from the candidate using dataset-specific templates: “This example is {}.” as the default, or the task-specific variants “The text expresses {}.”, “The customer’s message is about {}.”, “The topic of this news article is {}.”, and “The answer to the question is {}.”.

In the descriptions setting, the raw label string is replaced by a short descriptive phrase (for example, “world” $\rightarrow$ “world news: international politics, countries, governments, conflicts, diplomacy, and global events”), rather than concatenating the label and its description.

c) Rationale-assisted CSQA: To investigate whether the performance gap on CSQA is partly attributable to missing intermediate reasoning, we augment each question with an LLM-generated rationale before scoring. The candidate answers, scoring function, and decision rule remain unchanged. We evaluate three conditions: the original question (*Base*), the question paired with its corresponding rationale (*Rat*), and a control condition in which rationales are randomly reassigned across questions (*Shuf*). The shuffled condition helps distinguish improvements due to relevant reasoning content from effects caused simply by adding more text or increasing the input length.

In addition to accuracy, we report a standardized answer margin,

$$m_i = \frac{s_{i, \text{correct}} - \max_{k \neq \text{correct}} s_{i, k}}{\sigma_i + 10^{-9}},$$

where $s_{i, \text{correct}}$ is the score assigned to the correct answer, $\max_{k \neq \text{correct}} s_{i, k}$ is the highest score assigned to any incorrect candidate, and σ_i is the population standard deviation of all candidate scores for question i under the given model and condition. The reported margin is averaged across questions.

The sign and magnitude of m_i indicate how strongly the model separates the correct answer from its strongest competitor. A positive margin means that the correct answer receives a higher score than every incorrect candidate, with larger positive values indicating stronger separation. A margin near zero indicates that the correct answer and the strongest competing answer receive very similar scores. A negative margin means that at least one incorrect candidate is scored above the correct answer, with increasingly negative values indicating greater separation in favor of the incorrect candidate.

d) Training: Fine-tuning uses the cross-entropy objective of Section III.

- Bi-encoders: softmax over all K labels, so the label embeddings act as tied classifier weights.
- Cross-encoders: premise–hypothesis pairs under the same dataset template; the candidate set is the correct label plus up to 15 sampled negatives per example (all $K - 1$ when $K \leq 16$).
- Optimizer: AdamW, lr = 2×10^{-5} , weight decay 0.01, $\tau = 0.05$, gradient clipping at 1.0, 10% warmup.

Evaluation reports accuracy, and chance-normalized accuracy (acc – chance)/(1 – chance) when comparing across datasets.

Algorithm 1 Zero-shot / Fine-tuned Inference Pipeline

Require: Input text x , candidate set $\{c_1, \dots, c_K\}$, model type

```

1: if model is bi-encoder then
2:    $\mathbf{e}_x \leftarrow \text{norm}(\text{pool}(f_\theta(x)))$ 
3:   for  $k = 1 \dots K$  do
4:      $\mathbf{e}_k \leftarrow \text{norm}(\text{pool}(f_\theta(c_k)))$     ▷ Cached offline if
        set is fixed
5:      $s_k \leftarrow \mathbf{e}_x^\top \mathbf{e}_k$ 
6:   end for
7: else                                ▷ model is cross-encoder
8:   for  $k = 1 \dots K$  do
9:      $s_k \leftarrow g_\theta([x; h(c_k)])$       ▷ Joint entailment or
        relevance pass
10:  end for
11: end if
12: return  $\arg \max_k s_k$ 

```

V. IMPLEMENTATION

The bi-encoder wrapper below implements the scoring of Section III (prefixes for e5 are added to `texts` beforehand).

```

def encode(texts):
    batch = tokenizer(texts, padding=True, truncation=True,
                      max_length=512, return_tensors="pt").to(
        device)
    with torch.no_grad():
        h = model(**batch).last_hidden_state
        emb = h[:, 0] if pooling == "cls" else average_pool(h, batch["
        attention_mask"])
    return F.normalize(emb.float(), dim=1)

```

Fine-tuning adds the loss of Section III on top of this wrapper.

```

q = encode(queries)           # (B, d)
c = encode(candidates)        # (K, d), same encoder
logits = (q @ c.T) / temperature # cosine / tau
loss = F.cross_entropy(logits, labels)
loss.backward()
clip_grad_norm_(model.parameters(), 1.0)
optimizer.step(); scheduler.step()

```

The NLI cross-encoder follows Section III-D directly:

```

hypothesis = template.format(candidate)
inputs = tokenizer(text, hypothesis, padding=True,
                  truncation=True, max_length=512,
                  return_tensors="pt")
logits = model(**inputs).logits
score = logits[:, entailment_idx].float()

```

VI. EXPERIMENTAL SETUP

a) *Data and models:* We use four datasets spanning association to reasoning (Table VI):

- AG News [33]: four-way topical classification.
- Banking77 [34]: 77-way intent classification with many semantically adjacent labels.
- Emotion [35]: six-way affect recognition.
- CommonsenseQA [36]: five-way multiple choice with plausible distractors.

Table V illustrates the range of decisions across the datasets. AG News often exposes strong topical cues, whereas Banking77 requires discrimination among semantically adjacent intents. Emotion includes explicitly expressed affect, but many items require inferring the label from context. CSQA similarly ranges from questions with strong lexical associations to cases in which several options are plausible and the correct answer depends on a commonsense relation.

The explicit Emotion and easy CSQA examples are selected cases that both e5-small-v2 and DeBERTa-B-ZS answer correctly; they illustrate the easier end rather than typical items.

For AG News, Emotion, and Banking77, we fine-tune on the training split and evaluate on the test split. For CSQA, evaluation uses the validation split because the test labels are hidden, and the E5 CSQA fine-tune is trained on the CSQA training split. The models are listed in Tables I and II and Section III-E.

b) *Rationale subset:* The rationale experiment uses 300 questions drawn uniformly at random (seed 0) from the 1,221 CSQA validation items. Base accuracies in Table VIII differ slightly from Table III because they are computed on this 300-item subset rather than all 1,221 validation items. Rationales were generated with ChatGPT (GPT 5.6 Luna). The generator received only the question text, without answer options. The prompt requested 1–3 sentences describing the relevant commonsense facts without stating or guessing the answer, and included two synthetic demonstrations unrelated to CSQA.

We additionally report a filtered subset of 277 examples after excluding rationales with obvious lexical overlap with the correct option. This conservative lexical filter does not detect paraphrastic leakage.

c) *Baselines:* Chance is the lower baseline. The label-names-only zero-shot setting is the baseline for both the descriptions and the fine-tuning experiments, so every reported change is relative to the same model’s label-names accuracy.

d) *Hardware and Software:* Experiments were run on an NVIDIA RTX 3070 Ti Laptop GPU (8 GB) using PyTorch and Hugging Face Transformers.

VII. QUANTITATIVE RESULTS

Tables III–VII give accuracy with label names only, with class descriptions, and after fine-tuning. Table IX varies one design choice at a time.

TABLE III: Zero-shot accuracy (baseline) (%) with label names only. Shading shows chance-normalized accuracy; best in bold, second best underlined **among encoder models**. †Results for GPT-4o represent an external generative LLM baseline aggregated from zero/few-shot evaluations in VLDB, ACL Anthology (AveniBench), and CodeSOTA.

Model	Classification datasets				Commonsense QA
	AG News	Emotion	Banking77	Avg	CSQA
<i>Bi-encoders</i>					
bge-small-en-v1.5	69.3	49.6	67.9	62.3	40.4
bge-m3	73.1	41.7	64.7	59.8	42.4
all-MiniLM-L6-v2	64.2	43.7	62.3	56.7	40.3
e5-small-v2	68.5	46.4	58.9	57.9	41.1
<i>Cross-encoders</i>					
bge-reranker-base	64.4	38.4	55.6	52.8	35.3
DeBERTa-L-ZS	89.3	<u>70.0</u>	72.2	<u>77.2</u>	53.6
BART-large-MNLI	67.4	49.3	50.8	55.8	40.6
NLI-DeBERTa-B	66.9	35.8	33.1	45.3	43.3
bge-m3-ZS	86.8	69.8	65.3	74.0	34.0
DeBERTa-B-ZS	<u>87.8</u>	74.8	<u>71.2</u>	77.9	<u>43.8</u>
<i>Proprietary generative baselines</i>					
GPT-4o†	86.7	92.4	96.4	91.8	85.4
Chance	25.0	16.7	1.3	14.3	20.0

TABLE IV: Zero-shot accuracy (%) with class descriptions. Bracketed values are the change from label names only (Table III); teal marks gains, red marks losses. Best in bold. Avg is the raw mean over the three tasks. CSQA is omitted because its options have no class descriptions.

Model	AG News	Emotion	Banking77	Avg
<i>Bi-encoders</i>				
bge-m3	80.8 (+7.7)	44.2 (+2.5)	69.2 (+4.5)	64.7 (+4.9)
bge-small-en-v1.5	80.8 (+11.5)	46.7 (-2.9)	70.4 (+2.5)	66.0 (+3.7)
e5-small-v2	79.9 (+11.4)	46.2 (-0.2)	65.0 (+6.1)	63.7 (+5.8)
all-MiniLM-L6-v2	77.5 (+13.3)	45.1 (+1.4)	66.6 (+4.3)	63.1 (+6.3)
<i>Cross-encoders</i>				
bge-reranker-base	71.5 (+7.1)	46.7 (+8.3)	60.7 (+5.1)	59.6 (+6.8)
DeBERTa-L-ZS	89.1 (-0.2)	68.7 (-1.3)	67.4 (-4.8)	75.1 (-2.1)
BART-large-MNLI	78.9 (+11.5)	55.2 (+5.9)	44.0 (-6.8)	59.4 (+3.5)
NLI-DeBERTa-B	60.6 (-6.3)	41.2 (+5.4)	39.8 (+6.7)	47.2 (+1.9)
bge-m3-ZS	88.8 (+2.0)	68.4 (-1.4)	59.6 (-5.7)	72.3 (-1.7)
DeBERTa-B-ZS	89.6 (+1.8)	68.6 (-6.2)	63.5 (-7.7)	73.9 (-4.0)
Chance	25.0	16.7	1.3	14.3

a) *Label names only*: The bi-encoders average 56.7–62.3 across the classification datasets but only 40.3–42.4 on CSQA, about twice chance. The best cross-encoders (the zero-shot v2.0 models) reach 74.0–77.9 across the classification datasets, yet their CSQA accuracy varies from 34.0 to 53.6. The two rankings are therefore not tied: bge-m3-ZS is strong on classification (74.0) but below every bi-encoder on CSQA (34.0), while NLI-DeBERTa-B is lower on classification (45.3) but on par with or above the bi-encoders on CSQA (43.3).

Table IX shows that changing the training recipe moves classification accuracy by 32.6 points and CSQA by 0.5, whereas scaling the same recipe from base to large leaves classification accuracy nearly unchanged but adds 9.8 points on CSQA. The last row should be read as a model-family comparison rather than a provenance claim: we could not verify from the available model-card lineage

whether the zero-shot v2.0 checkpoint was trained directly from the base bge-m3 weights.

b) *Class descriptions*: Descriptions improve seven of ten models, by +1.9 to +6.8 on the classification average, with the largest gains on AG News for the bi-encoders (up to +13.3). They hurt the three zero-shot v2.0 models (−1.7 to −4.0 average), mostly on Banking77 (−4.8 to −7.7). CSQA has no class descriptions, so this setting cannot be applied to it. Retrieving external sentences as pseudo-descriptions for label prompts has been shown to improve zero-shot accuracy, for example raising Sentence-T5 on AG News from 70.4% to 76.6% [4]; the same work reports 55.5% on CommonsenseQA with retrieval.

c) *Fine-tuning*: Fine-tuning improves every classification-dataset cell, by +4.0 to +32.6 points, removing 29–58% of the remaining error on average. On CSQA the bi-encoders change by −0.3 and +0.8

TABLE V: Evaluation examples

Dataset	Input	Options	Correct
AG News	“Allianz to fight US court ruling on WTC attacks MUNICH - German insurance concern Allianz said on Tuesday it would fight a US jury decision...”	World, Sports, Business, Sci/Tech	Business
Banking77	“My ATM transaction was unsuccessful.”	cash deposit , declined cash withdrawal, ... (77 intent options)	declined cash withdrawal
Emotion (explicit)	“i wonder how it feels to be loved by someone you love”	sadness, joy, love, anger, fear, surprise	love
Emotion (implicit)	“i am finally starting to feel like i have a real life here in san vicente and i am no longer on a strange confusing extended vacation”	sadness, joy, love, anger, fear, surprise	fear
CSQA (easy)	“Farm land makes use of what?”	countryside, michigan, north dakota, farming areas, illinois	farming areas
CSQA (distractors)	“The fox walked from the city into the forest, what was it looking for?”	pretty flowers, hen house, natural habitat , storybook, dense forest	natural habitat

TABLE VI: Eval datasets. K : # of labels (answer options for CSQA).

Dataset	K	Chance	Eval size	Demand
AG News	4	25.0	7,600	Topic cues
Banking77	77	1.3	3,080	Near-identical intents
Emotion	6	16.7	2,000	Implicit affect
CSQA	5	20.0	1,221	Plausible distractors

(MiniLM, bge-small, within sampling noise of roughly ± 3 points) and +6.0 (e5), while the cross-encoders gain +12.6 and +9.9, removing 17–22% of their remaining error. The best fine-tuned CSQA accuracy is 56.4%, against 84.1% on the classification average for the same model (chance-normalized: 45.5% vs. 82.0%).

d) Rationale-assisted CSQA: Table VIII reports Base, Rat, and Shuf accuracy, the accuracy change from adding the rationale (on all questions and after filtering), and the standardized answer margin. The accuracy point estimate increases for all ten scorers: by 5.0–10.3 points for bi-encoders and 3.0–8.0 for cross-encoders (5.3 for the reranker). After filtering, gains remain positive (3.6–9.4 and 1.4–6.1 points), although only three filtered comparisons are individually significant at unadjusted $p < 0.05$. We therefore interpret the cross-model consistency as suggestive rather than as evidence of a significant effect for every model.

Shuffled rationales instead lower accuracy to 25.0–36.7%, below Base for all ten scorers, so the gain does not come from extra text alone. The margin tells the same story: it is negative for every scorer under Base, becomes less negative with relevant rationales, and turns positive only for DeBERTa-L-ZS ($-0.12 \rightarrow +0.14$). Making the missing relation explicit thus improves answer separability without changing the scorer, candidate set, or decision rule.

VIII. ANALYSIS AND DISCUSSION

a) Where label cues often suffice: On AG News and Banking77, label names or descriptions often carry much

of the information needed for classification. Emotion also includes cases requiring contextual inference; after fine-tuning, it has the lowest chance-normalized accuracy for four of the five models.

b) Constructing versus recognizing an inference: The rationale experiment sharpens the interpretation of the CSQA gap. With the original question alone, the correct answer is often poorly separated from plausible distractors: the mean standardized answer margin is negative for every tested scorer. Once a relevant rationale makes the missing commonsense relation explicit, both accuracy and standardized answer margins improve consistently across model families. The strongest cross-encoder even shifts from a negative to a positive mean standardized answer margin.

This suggests that part of the difficulty lies not in recognizing the correct answer once the relevant relation is represented, but in *constructing* that intermediate relation from the original question. Fixed-set scorers can exploit information that is already present in their input, but their scoring objective does not itself provide a mechanism for explicitly generating missing intermediate facts.

The shuffled-rationale control supports this interpretation. Unrelated rationales do not provide the same benefit and generally reduce accuracy, showing that the gain is not explained by additional context alone. What matters is whether the added text exposes a relation that makes the correct candidate more semantically distinguishable from the distractors.

The rationale generator was given neither the answer options nor any indication of the correct answer, so the generated explanation could not be directly tailored to a particular candidate. However, a rationale may still indirectly narrow the plausible answer space by introducing semantically informative cues. Our filtering procedure therefore removes direct lexical overlap, but cannot eliminate all forms of indirect candidate-specific information.

c) Implications for bounded decision models: These results also qualify the motivating comparison with

TABLE VII: Fine-tuned model accuracy (%). Parentheses show the accuracy-point change from the label-names-only zero-shot model (Table III); Avg is over AG News, Emotion, and Banking77.

Model	AG News	Emotion	Banking77	Avg	CSQA
<i>Fine-tuned bi-encoders</i>					
all-MiniLM-L6-v2	88.5 (+24.3)	60.8 (+17.1)	68.4 (+6.1)	72.6 (+15.8)	40.0 (-0.3)
bge-small-en-v1.5	91.0 (+21.7)	65.1 (+15.5)	74.3 (+6.4)	76.8 (+14.5)	41.2 (+0.8)
e5-small-v2	91.4 (+22.9)	67.6 (+21.2)	68.3 (+9.4)	75.8 (+17.8)	47.1 (+6.0)
<i>Fine-tuned cross-encoders</i>					
DeBERTa-B-ZS	91.8 (+4.0)	83.6 (+8.8)	77.0 (+5.8)	84.1 (+6.2)	56.4 (+12.6)
NLI-DeBERTa-B	91.1 (+24.2)	68.4 (+32.6)	65.7 (+32.6)	75.1 (+29.8)	53.2 (+9.9)
Chance	25.0	16.7	1.3	14.3	20.0

TABLE VIII: Rationale-assisted CSQA on the 300-question subset (accuracy in %).

Model	Accuracy			Δ Rat-Base		Std. correct margin	
	Base	Rat	Shuf	All	Filtered	Base	Rat
<i>Bi-encoders</i>							
all-MiniLM-L6-v2	40.3	49.3	30.7	+9.0	+7.9 [†]	-0.47	-0.20
e5-small-v2	38.7	49.0	31.3	+10.3	+9.4 [†]	-0.48	-0.12
bge-small-en	38.3	46.3	25.0	+8.0	+6.9 [†]	-0.46	-0.21
bge-m3	44.3	49.3	32.3	+5.0	+3.6	-0.38	-0.22
<i>Cross-encoders</i>							
DeBERTa-L-ZS	52.3	55.3	36.7	+3.0	+1.4	-0.12	+0.14
DeBERTa-B-ZS	45.3	53.3	36.3	+8.0	+6.1	-0.37	-0.04
BART-large-MNLI	38.3	42.0	35.7	+3.7	+1.4	-0.54	-0.45
NLI-DeBERTa-B	42.7	48.3	34.7	+5.7	+3.6	-0.47	-0.23
bge-m3-ZS	33.3	40.3	29.7	+7.0	+5.4	-0.60	-0.43
<i>Reranker</i>							
bge-reranker-b	33.3	38.7	25.3	+5.3	+3.6	-0.67	-0.50

[†] Unadjusted exact two-sided McNemar $p < 0.05$ for Rat vs. Base on Filtered; exploratory, with no correction for ten comparisons.

TABLE IX: Controlled comparisons: change in classification average / CSQA accuracy (points) when one factor changes.

Change	Δ Classif.	Δ CSQA
Recipe: NLI-DeBERTa-B $\rightarrow$ DeBERTa-B-ZS	+32.6	+0.5
Scale: DeBERTa-B-ZS $\rightarrow$ DeBERTa-L-ZS	-0.7	+9.8
Backbone: DeBERTa-L-ZS $\rightarrow$ bge-m3-ZS	-3.2	-19.6
Bi-encoder $\rightarrow$ NLI model on bge-m3	+14.2	-8.4

Laya/JEV-style bounded decision models. Strong performance on a fixed candidate set does not by itself imply general reasoning ability: a bounded scorer may perform very well whenever the information needed for the decision is already encoded in the input. The rationale experiment demonstrates exactly this case, since the same scorers become substantially stronger once the missing relation is supplied externally.

This leaves an important boundary. Single-pass decision models may be sufficient for many tasks in which the relevant evidence is already semantically available, even when the label space is large or fine-grained. Generative or iterative computation becomes more useful when the decision depends on intermediate information that must first be constructed rather than merely recognized.

d) Sensitivity: Results depend strongly on input conventions, as the prefix experiment in Section IV shows. Descriptions also help or hurt depending on the model

and dataset, with opposite signs for BART-large-MNLI and NLI-DeBERTa-B on different datasets, so their effect depends on wording and on how closely the input matches the training format.

e) Computation: Parameters range from 22M (MiniLM) to 568M (bge-m3). As derived in Section III, a cross-encoder needs K passes per text (77 on Banking77). We measured per-query latency only for the cross-encoders; peak GPU memory and bi-encoder latency were not recorded. For example, the DeBERTa-v3-base zero-shot model takes about 14–16 ms per query on Emotion and AG News and roughly 121–141 ms on Banking77. Bi-encoders should be much cheaper once label embeddings are cached, but we did not measure this.

f) Limitations: Each configuration is a single run, so differences below about 1–3 points (depending on dataset size) should not be interpreted. Rationales come from a single non-deterministic generator. Only five models were fine-tuned, the zero-shot v2.0 training data may overlap with the evaluation datasets, and our conclusions about reasoning rest on one probe and on label formatting choices for CSQA. The method is unlikely to work when answers require multi-step inference over unstated facts, which is exactly what the CSQA results show.

IX. CONCLUSION

We compared bi-encoders, NLI cross-encoders, and a reranker as zero-shot classifiers on three classification datasets and CommonsenseQA. Class descriptions and fine-tuning improve the classification datasets substantially, although descriptions can hurt models already trained on label-as-hypothesis data. On CSQA, accuracy is largely unrelated to classification accuracy, fine-tuning barely helps bi-encoders, and cross-encoders gain more but stay well below their classification average. This suggests that similarity and entailment scoring capture semantic association rather than inference. The rationale-assisted CSQA experiment further separates two capabilities that are often conflated: *constructing* an inference and *recognizing* its consequence. When the missing commonsense relation is supplied explicitly, the same fixed-set scorers improve in both accuracy and standardized answer margin, indicating that some failures arise before the final candidate-scoring step.

These results suggest a practical boundary for bounded decision models. Single-pass scorers can be sufficient when the information required for the decision is already semantically available in the input. Generative or iterative computation becomes increasingly valuable when the model must construct missing intermediate information before the correct candidate can be distinguished.

REFERENCES

- [1] TypeSafe AI, “Jev: A system-one decision model,” Product release, 2026, accessed: 2026-10-05.
- [2] N. M. and Convai Innovations, “Laya: An open-weight non-autoregressive decision model,” Hugging Face model card, 2026, accessed: 2026-10-05. [Online]. Available: <https://huggingface.co/convaiinnovations/laya>
- [3] N. Reimers and I. Gurevych, “Sentence-bert: Sentence embeddings using siamese bert-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.
- [4] J. Hong, J. Park, D. Kim, S. Choi, B. Son, and J. Kang, “Empowering sentence encoders with prompting and label retrieval for zero-shot text classification,” in *arXiv preprint arXiv:2212.10391v2*, 2023.
- [5] W. Yin, J. Hay, and D. Roth, “Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3914–3923.
- [6] M. Laurer, W. van Attevelde, A. Casas, and K. Welbers, “Less annotating, more classifying: Addressing the data scarcity bottleneck of text classification with nli models,” in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022.
- [7] e. a. Chai, “Description based text classification,” *arXiv preprint*, 2020.
- [8] e. a. Ma, “Issues with entailment-based zero-shot text classification,” in *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2021.
- [9] R. Nogueira and K. Cho, “Passage re-ranking with bert,” *arXiv preprint arXiv:1901.04085*, 2019.
- [10] S. Humeau, K. Shuster, M.-A. Lachaux, and J. Weston, “Poly-encoders: Architectures and pre-training strategies for fast and accurate multi-sentence scoring,” in *International Conference on Learning Representations*, 2020.
- [11] O. Khattab and M. Zaharia, “Colbert: Efficient and effective passage search via contextualized late interaction over bert,” in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 39–48.
- [12] N. Thakur, N. Reimers, J. Daxenberger, and I. Gurevych, “Augmented sbert: Data augmentation method for improving bi-encoders for pairwise sentence scoring tasks,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 296–310.
- [13] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, “Chain-of-thought prompting elicits reasoning in large language models,” 2023. [Online]. Available: <https://arxiv.org/abs/2201.11903>
- [14] S. Hao, S. Sukhbaatar, D. Su, X. Li, Z. Hu, J. Weston, and Y. Tian, “Training large language models to reason in a continuous latent space,” 2026. [Online]. Available: <https://arxiv.org/abs/2412.06769>
- [15] Y. Koishekenov, A. Lipani, and N. Cancedda, “Encode, think, decode: Scaling test-time reasoning with recursive latent thoughts,” 2025. [Online]. Available: <https://arxiv.org/abs/2510.07358>
- [16] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, arXiv:2002.10957.
- [17] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei, “Text embeddings by weakly-supervised contrastive pre-training,” *arXiv preprint arXiv:2212.03533*, 2022.
- [18] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, “C-Pack: Packed resources for general chinese embeddings,” *arXiv preprint arXiv:2309.07597*, 2023.
- [19] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *arXiv preprint arXiv:2402.03216*, 2024.
- [20] Sentence-Transformers, “all-MiniLM-L6-v2,” <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>, hugging Face model card.
- [21] intfloat, “e5-small-v2,” <https://huggingface.co/intfloat/e5-small-v2>, hugging Face model card.
- [22] BAAI, “bge-small-en-v1.5,” <https://huggingface.co/BAAI/bge-small-en-v1.5>, hugging Face model card.
- [23] —, “bge-m3,” <https://huggingface.co/BAAI/bge-m3>, hugging Face model card.
- [24] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” *arXiv preprint arXiv:2111.09543*, 2021.
- [25] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, “BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [26] Facebook AI, “bart-large-mnli,” <https://huggingface.co/facebook/bart-large-mnli>, hugging Face model card.
- [27] M. Laurer, “deberta-v3-large-zeroshot-v2.0,” <https://huggingface.co/MoritzLaurer/deberta-v3-large-zeroshot-v2.0>, hugging Face model card.
- [28] —, “deberta-v3-base-zeroshot-v2.0,” <https://huggingface.co/MoritzLaurer/deberta-v3-base-zeroshot-v2.0>, hugging Face model card.
- [29] Cross-Encoder, “nli-deberta-v3-base,” <https://huggingface.co/cross-encoder/nli-deberta-v3-base>, hugging Face model card.
- [30] M. Laurer, “bge-m3-zeroshot-v2.0,” <https://huggingface.co/MoritzLaurer/bge-m3-zeroshot-v2.0>, hugging Face model card.
- [31] BAAI, “bge-reranker-base,” <https://huggingface.co/BAAI/bge-reranker-base>, hugging Face model card.
- [32] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoy-

- anov, “Unsupervised cross-lingual representation learning at scale,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [33] X. Zhang, J. Zhao, and Y. LeCun, “Character-level convolutional networks for text classification,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [34] I. Casanueva, T. Temčinas, D. Gerz, M. Henderson, and I. Vulić, “Efficient intent detection with dual sentence encoders,” in *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, 2020.
- [35] E. Saravia, H.-C. T. Liu, Y.-H. Huang, J. Wu, and Y.-S. Chen, “CARER: Contextualized affect representations for emotion recognition,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
- [36] A. Talmor, J. Herzig, N. Lourie, and J. Berant, “Commonsenseqa: A question answering challenge targeting commonsense knowledge,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 4149–4158.